Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. We trained each model for 100 epochs with a batch size of 64.

2 Results

Figure 1 shows the test accuracy of Transformer by dataset.

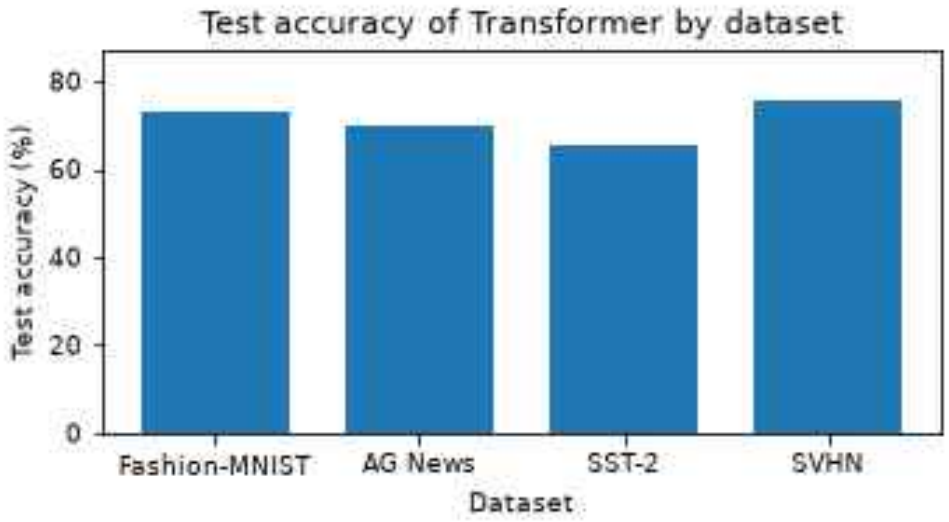


Figure 1: Test accuracy of Transformer by dataset for the models in this experiment.

Transformer scores 9.9 points higher on SVHN than on SST-2. Transformer reaches 65.7% test accuracy on SST-2 (Figure 1). As shown in Figure 1, Transformer scores 2.5 points higher on SVHN than on Fashion-MNIST. Transformer scores 21.3 points higher on AG News than on SST-2 (Figure 1).

Figure 2 shows the test accuracy during training.

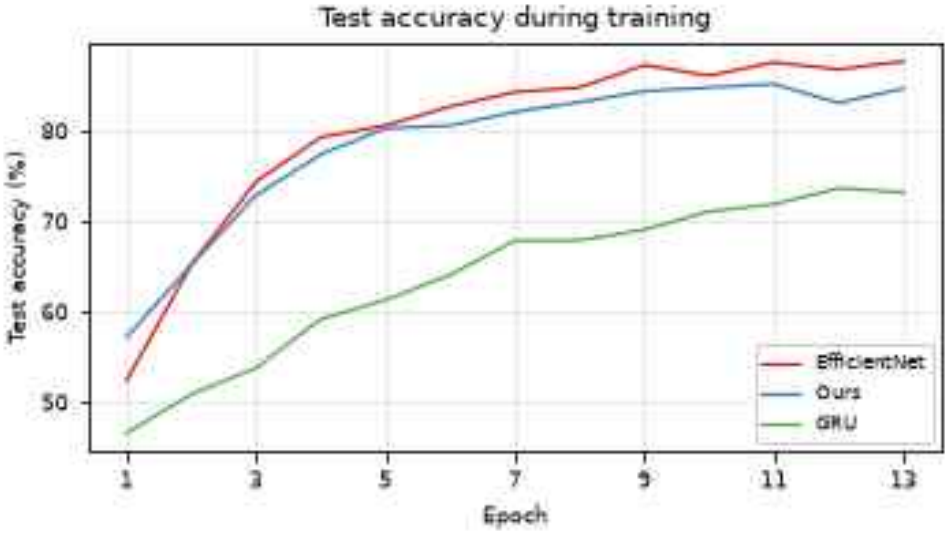


Figure 2: Test accuracy during training for the models in this experiment.

At epoch 13, GRU achieves 73.2% test accuracy. EfficientNet reaches about 70% test accuracy at epoch 3 (Figure 2). Ours reaches 85% test accuracy at epoch 13 (Figure 2). From epoch 2 to epoch 12, the test accuracy of GRU improves by a relative 27.0%.

Figure 3 shows the training loss.

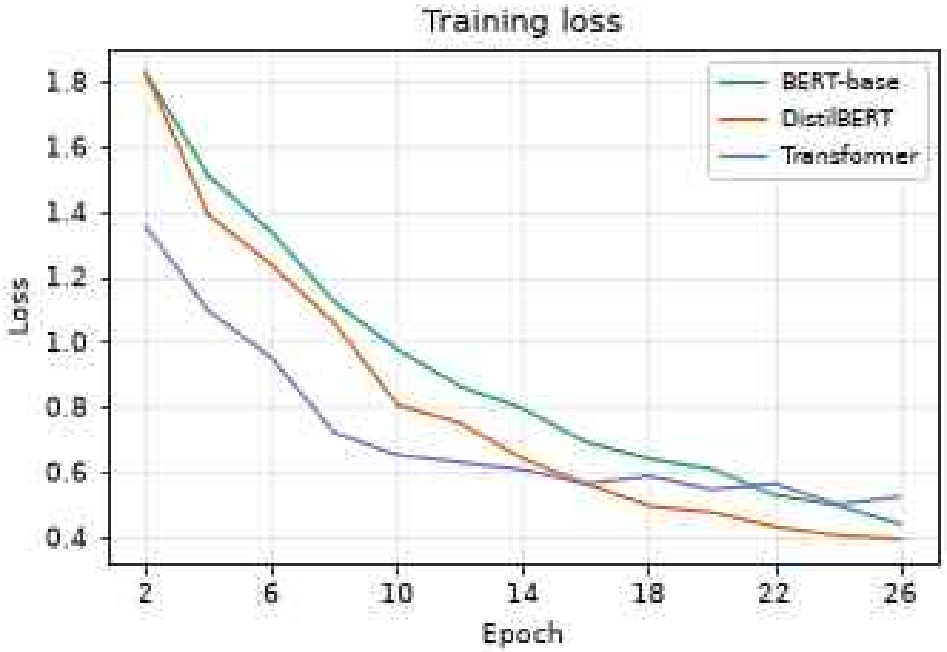


Figure 3: Training loss for the models in this experiment.

Transformer has a loss of 1.337 at epoch 6. DistilBERT has a loss of 0.609 at epoch 14 (Figure 3). At epoch 8, the loss of Transformer is 0.563 (Figure 3).

Figure 4 shows the test accuracy by dataset.

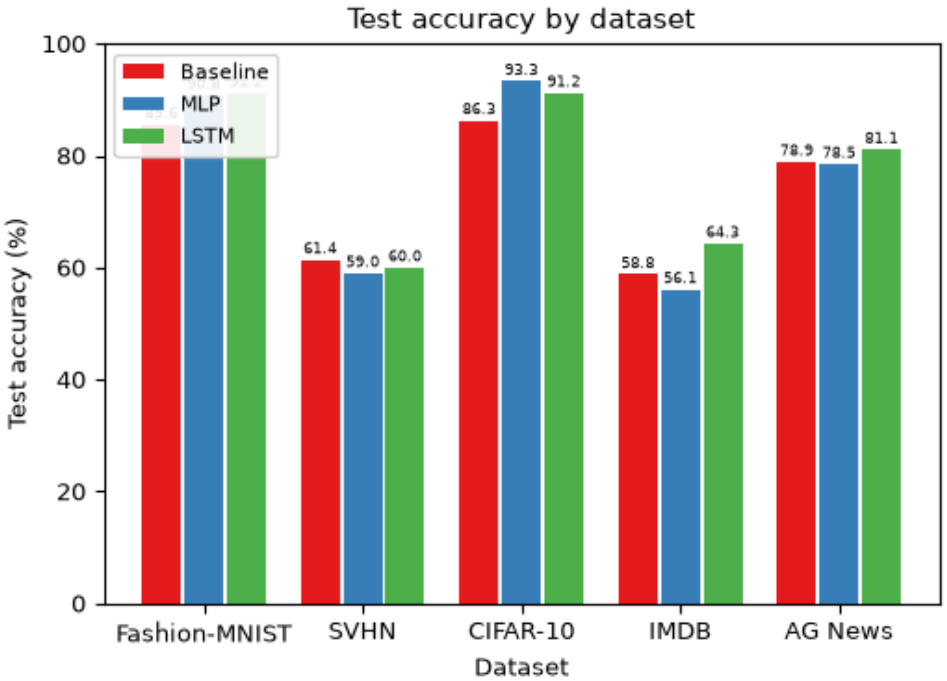


Figure 4: Test accuracy by dataset for the models in this experiment.

As shown in Figure 4, on IMDB, Baseline obtains a test accuracy of 63.8%. As shown in Figure 4, on SVHN, LSTM obtains a test accuracy of 50.0%. As shown in Figure 4, LSTM reaches 64% test accuracy on IMDB. MLP achieves 93.3% test accuracy on CIFAR-10 (Figure 4).

3 Discussion

XGBoost reaches 67.8% accuracy on STL-10. Overall, the results are in line with what we expected from the literature.